



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

PROJECT TITLE

Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment

A Software Engineering Project Submitted
By

Semester: Summer_21_22		Section:	Group Number:	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
01	AFNAN BIN ISLAM NAHIN	22-49350-3	25%	
02	AMAN ULLAH	22-49345-3	25%	
03	HRITTIK BHATTACHARJEE	21-44650-1	16.67%	
04	MD. HACHIBUR RAHMAN	22-46426-1	16.67%	
05	WALID BIN KHALED	22-46101-1	16.67%	

The project will be Evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]

Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	
CO4: <i>Develop</i> project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: AFNAN BIN ISLAM NAHIN Student ID: 22-49350-3 Contribution in Percentage (25%): <u>Contribution in the Project:</u> ▪ Functional Requirements ▪ Use case Diagram _____ Signature of the Student
Student Name: AMAN ULLAH Student ID: 22-49345-3 Contribution in Percentage (25%): <u>Contribution in the Project:</u> ▪ Project Idea & proposal ▪ Activity Diagram _____ Signature of the Student
Student Name: HRITTIK BHATTACHARJEE Student ID: 21-44650-1 Contribution in Percentage (16.67%): <u>Contribution in the Project:</u> ▪ Non-Functional Requirements ▪ Sequence Diagram _____ Signature of the Student

Student Name: MD.HACHIBUR RAHMAN

Student ID: 22-46426-1

Contribution in Percentage (16.67%):

Contribution in the Project:

- Non-Functional Requirements
- ER- Diagram

Signature of the Student

Student Name: WALID BIN KHALED

Student ID: 22-46101-1

Contribution in Percentage (16.67%):

Contribution in the Project:

- Non-Functional Requirements
- Class Diagram

Signature of the Student

1. PROJECT PROPOSAL

Title: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment

The Virtual Classroom Platform is a dynamic, interactive online learning solution designed to enhance remote education for students and instructors. In an era where virtual learning is essential, this platform offers a comprehensive suite of tools for real-time communication, content sharing, and student engagement. Core features include high-quality video conferencing, live chat, a digital whiteboard, breakout rooms, and interactive assessments like polls and quizzes. Additionally, the platform provides analytics and tracking dashboards, enabling instructors to monitor attendance, engagement, and academic progress.

With an emphasis on accessibility, data security, and regulatory compliance, this platform is designed for educational institutions, corporate training, and individual instructors. Scalable cloud infrastructure and cross-platform compatibility ensure reliability on various devices, from desktops to mobile. By facilitating a seamless, interactive, and secure learning experience, the Virtual Classroom Platform aims to meet the evolving needs of digital education and support active, engaging learning in a virtual environment.

1.1 Background to the Problem

1. Limited Interaction and Engagement

- **Problem:** Many virtual platforms struggle to replicate the interactive dynamics of a physical classroom, resulting in reduced student engagement and passive learning.
- **Impact:** Decreased participation, lack of real-time feedback, and potential loss of interest.

2. Technical Issues and Accessibility Barriers

- **Problem:** Students and educators often face technical challenges such as connectivity issues, device compatibility, and software glitches.
- **Impact:** Disruptions in learning, frustration, and a decrease in learning efficiency.

3. High Dependency on Internet Connectivity

- **Problem:** A reliable internet connection is crucial, yet it's not accessible to all students and educators, particularly in remote or under-resourced areas.
- **Impact:** Limited access to learning resources, interruptions in classes, and unequal opportunities for students.

4. Lack of a Unified User Experience Across Devices

- **Problem:** Platforms sometimes lack consistency in user experience across different devices (mobile, tablet, desktop), which can lead to confusion and frustration.
- **Impact:** Students and teachers face challenges navigating the platform, resulting in a less effective learning environment.

1.2 Solution to the Problem

1. Limited Interaction and Engagement

- **Solution:** Introduce interactive features like real-time polling, quizzes, digital whiteboards, and breakout rooms to increase engagement. Include reaction emojis, hand-raising, and a live chat to allow students to participate actively. Gamification, like awarding points for participation, can also encourage interaction.

2. Technical Issues and Accessibility Barriers

- **Solution:** Optimize the platform for low-bandwidth environments and include device compatibility testing for multiple operating systems and devices. Provide offline access to learning materials and downloadable lecture recordings to help students continue learning even with connectivity issues. Regular updates and a dedicated support team can also help address glitches and device compatibility issues.

3. High Dependency on Internet Connectivity

- **Solution:** Offer offline capabilities where students can download lectures, resources, and assignments. Optimize the platform for low-bandwidth use and introduce a text-based chat mode for low connectivity situations. For areas with very limited access, provide downloadable resources and recorded lessons that can be viewed without internet.

4. Lack of a Unified User Experience Across Devices

- **Solution:** Implement responsive design principles to ensure a consistent experience on all devices, including desktops, tablets, and mobile phones. Conduct extensive testing across devices and provide a simplified interface that automatically adjusts to different screen sizes, making navigation easier and reducing user frustration.

Objective

The objective of the virtual classroom platform is to create an engaging and interactive online learning environment that enhances educational experiences for students and educators alike.

1. **Facilitate Seamless Learning:** Provide a user-friendly interface that enables students to access course materials, submit assignments, and participate in discussions easily, promoting a seamless learning experience.
2. **Enhance Engagement and Interaction:** Incorporate interactive tools such as live quizzes, polls, and breakout rooms to foster collaboration and engagement among students, helping them stay motivated and connected.
3. **Support Personalized Learning:** Implement adaptive learning technologies that cater to individual learning styles and paces, allowing students to thrive in a personalized educational setting.
4. **Strengthen Community Building:** Create a supportive online community where students and educators can collaborate, share resources, and offer peer support, enhancing the overall educational experience.

Functionality:

1. Live Video and Audio Conferencing
2. Screen Sharing
3. Text Chat and Messaging
4. Content Sharing and Resource Library
5. Attendance Tracking
6. Polls, Quizzes, and Assessments
7. Session Recording and Playback
8. Analytics and Reporting
9. Push Notifications and Reminders
10. Privacy and Security Features

Target Audience and how different groups can benefit by Virtual Classroom Platform:

1. Students

Benefits:

- **Accessibility:** Offers a flexible learning environment accessible from anywhere.
- **Collaboration:** Provides tools for group projects and discussions, enhancing peer interaction.
- **Resource Sharing:** Allows easy access to class materials, assignments, and resources.

2. Teachers/Educators

Benefits:

- **Streamlined Teaching:** Facilitates lesson planning and delivery through organized modules.
- **Engagement Tools:** Incorporates interactive features like quizzes, polls, and forums to keep students engaged.
- **Assessment Tracking:** Enables efficient tracking of student performance and progress through analytics.

3. Parents

Benefits:

- **Involvement:** Offers insights into their child's learning journey and academic progress.
- **Communication:** Facilitates direct communication with teachers through messaging features.
- **Resource Access:** Provides access to educational resources to support learning at home.

4. School Administrators

Benefits:

- **Management Tools:** Simplifies administrative tasks, such as enrollment and attendance tracking.
- **Data Insights:** Offers analytics to help improve curriculum and teaching methods based on student performance.
- **Scalability:** Provides a platform that can grow with the institution, accommodating more students and courses as needed.

5. Educational Institutions (Schools/Colleges)

Benefits:

- **Cost-Effectiveness:** Reduces the need for physical infrastructure and materials, leading to cost savings.
- **Curriculum Development:** Facilitates the creation and delivery of diverse courses, reaching a wider audience.

- **Brand Enhancement:** Enhances institutional reputation by adopting modern educational technologies.

6. Tutoring Services

Benefits:

- **Wider Reach:** Allows tutors to connect with students beyond geographical limitations.
- **Resource Integration:** Enables the sharing of learning materials and sessions through a centralized platform.
- **Customizable Learning Paths:** Offers personalized tutoring options tailored to individual student needs.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model:

Selected Process Model: Agile Methodology (Scrum Framework)

The **Agile methodology**, specifically the **Scrum framework**, is selected for developing the Virtual Classroom Platform. Agile emphasizes iterative development, frequent feedback, and adaptability to change, making it highly suitable for the dynamic nature of the platform.

Arguments for Selecting Agile (Scrum)

1. Flexibility in Requirements:

- Educational needs evolve rapidly, especially with technological advancements. Agile allows frequent reassessment of requirements and easy integration of new features.

2. User-Centered Development:

- Scrum involves regular feedback from stakeholders (e.g., students, teachers, administrators) at the end of each sprint. This ensures the platform aligns with user expectations and delivers high usability.

3. Iterative Delivery:

- Agile's sprint-based delivery ensures small, functional components (e.g., login system, video conferencing) are developed and tested incrementally. This reduces the risk of large-scale failures at the end.

4. Focus on Quality:

- Continuous testing and review in Agile minimize bugs and ensure that each feature is thoroughly validated before release.
- Example: Testing the video conferencing module in multiple sprints ensures high-quality real-time communication.

5. Collaboration:

- Scrum's daily standups and collaborative approach foster seamless communication between developers, designers, testers, and stakeholders, ensuring alignment with project goals.

6. Time Efficiency:

- Delivering functional increments in each sprint allows institutions to start using parts of the platform (e.g., assignment submission) even before the entire system is complete.

Evidence to Support Agile Selection

1. Industry Precedent:

- Agile is widely used in the development of dynamic, user-driven systems like e-learning platforms, mobile apps, and collaborative tools. Examples include platforms like Google Classroom and Microsoft Teams.

2. Complexity Handling:

- Virtual Classroom Platforms require the integration of multiple complex subsystems (e.g., live video streaming, data encryption, cloud storage). Agile's modular approach simplifies the development and testing of these components.

3. User Feedback in Real-Time:

- Incorporating teacher and student feedback early ensures that features like screen sharing or breakout rooms work effectively and meet expectations.

4. Risk Mitigation:

- Incremental delivery ensures that potential risks (e.g., scalability issues) are identified and addressed early in development.

2.2 Project Role Identification and Responsibilities

The successful development of the Virtual Classroom Platform requires collaboration between multiple roles and stakeholders. Below is an identification of key roles and their responsibilities:

1. Project Manager

Responsibilities:

Plan, execute, and monitor the project to ensure it is completed on time, within scope, and within budget. Manage risks and resolve conflicts to ensure project success.

2. Product Owner

Responsibilities:

Gather and prioritize requirements from stakeholders (e.g., educators, students, institutions). Maintain the Product Backlog by defining user stories and acceptance criteria. Ensure the development team focuses on features that add the most value to users.

3. Development Team

This includes Software Developers, Front-End Engineers, and Back-End Engineers.

Responsibilities:

Design, develop, and implement the platform's features and functionalities.

4. UI/UX Designer

Responsibilities:

Design intuitive and visually appealing user interfaces for the platform. Conduct user research to ensure the platform is accessible and easy to use for a diverse audience. Create wireframes, prototypes, and design mockups for the platform's various components (e.g., dashboard, live class interface).

5. Quality Assurance (QA) Engineer

Responsibilities:

Develop test cases to validate the platform's features and functionalities.

Perform functional, performance, and security testing to ensure the platform meets requirements.

Report bugs and work closely with developers to resolve issues.

6. Database Administrator (DBA)

Responsibilities:

Design and manage the database schema to store user data, course content, and activity logs securely. Optimize database performance and ensure data integrity. Implement backup and recovery procedures to prevent data loss.

7. Stakeholders (Educators, Students, Administrators)

Responsibilities:

Provide feedback on platform requirements, usability, and features. Participate in testing to ensure the platform meets real-world needs. Act as end-users who adopt and promote the platform within their institution.

8. Scrum Master (Specific to Agile Projects)

Responsibilities:

Facilitate Agile ceremonies (e.g., sprint planning, daily standups, retrospectives). Remove obstacles that block the team's progress. Ensure the team adheres to Agile principles and practices. Foster collaboration among team members to achieve sprint goals.

9. End Users (Teachers, Students, and Administrators)

Responsibilities:

Use the platform for its intended purposes, such as hosting classes, attending lectures, or managing administrative tasks. Provide feedback to improve the platform's features and usability.

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marks Distribution (Maximum 3X5=15)					
Marking Criteria	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					